

Step-by-Step Derivative

$$f(x) = \frac{2}{x^2}$$

Step 1: Constant Multiple Rule

$$\frac{d}{dx} \left[\frac{2}{x^2} \right] = -\frac{4}{x^3}$$

Step 2: Power Rule

$$\frac{d}{dx} \left[\frac{1}{x^2} \right] = -\frac{2}{x^3}$$

Step 3: Simplify

$$\frac{d}{dx} \left[-\frac{4}{x^3} \right] = -\frac{4}{x^3}$$

Final Result

$$f'(x) = -\frac{4}{x^3}$$